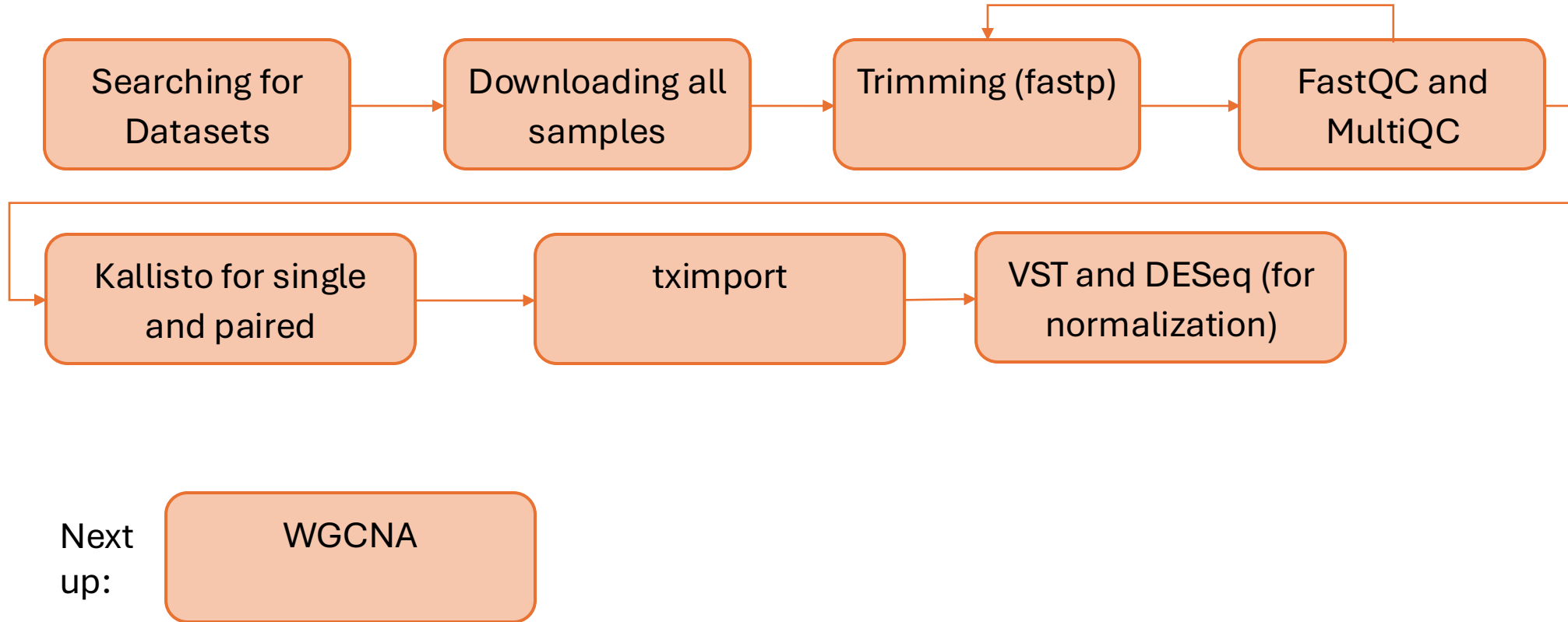


Current progress



Datasets

Biotrophic pathogens:

- *Golovinomyces orontii*
- *Erysiphe cichoracearum*
- *Hyaloperonospora arabidopsidis* (2 Datasets)
- *Plasmodiophora brassicae* (2 Datasets)
- Turnip mosaic virus (TuMV) (2 Datasets)
- *Heterodera schachtii*
- Turnip yellows virus (TuYV)
- Cauliflower mosaic virus (CaMV)
- *Meloidogyne javanica*

Control:

- Drought
- *Botrytis cinerea* (necrotrophic)
- *Alternaria brassicae* (necrotrophic)

Altogether:

283 samples

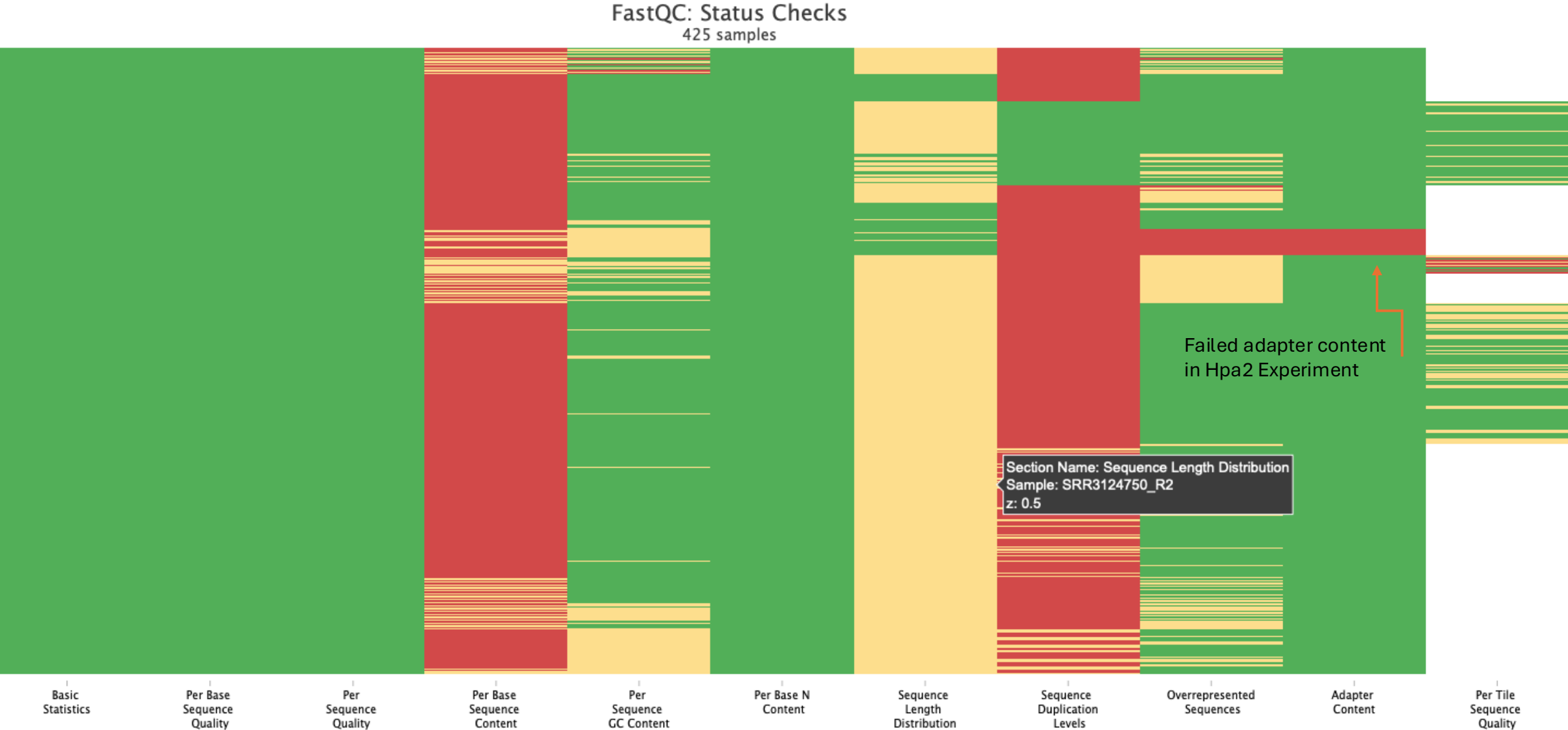
141 single-end, 142 paired-end

<https://gitlab.lrz.de/00000000014B0D03/bachelor-thesis-luisa/-/tree/main/data>

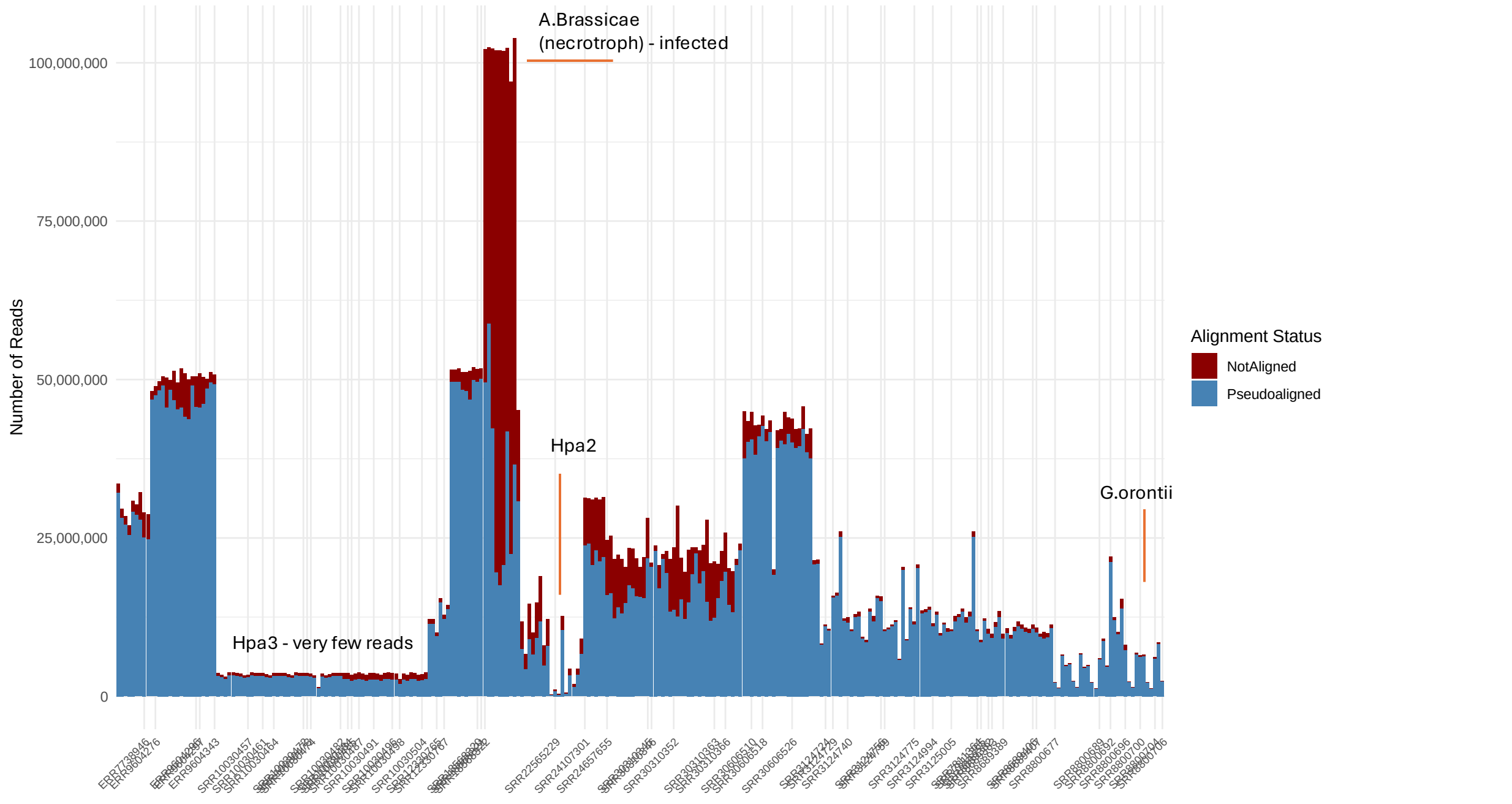
MultiQC after basic trimming with fastp



Harder trimming on P.brassicae samples -> change quality threshold to improve quality

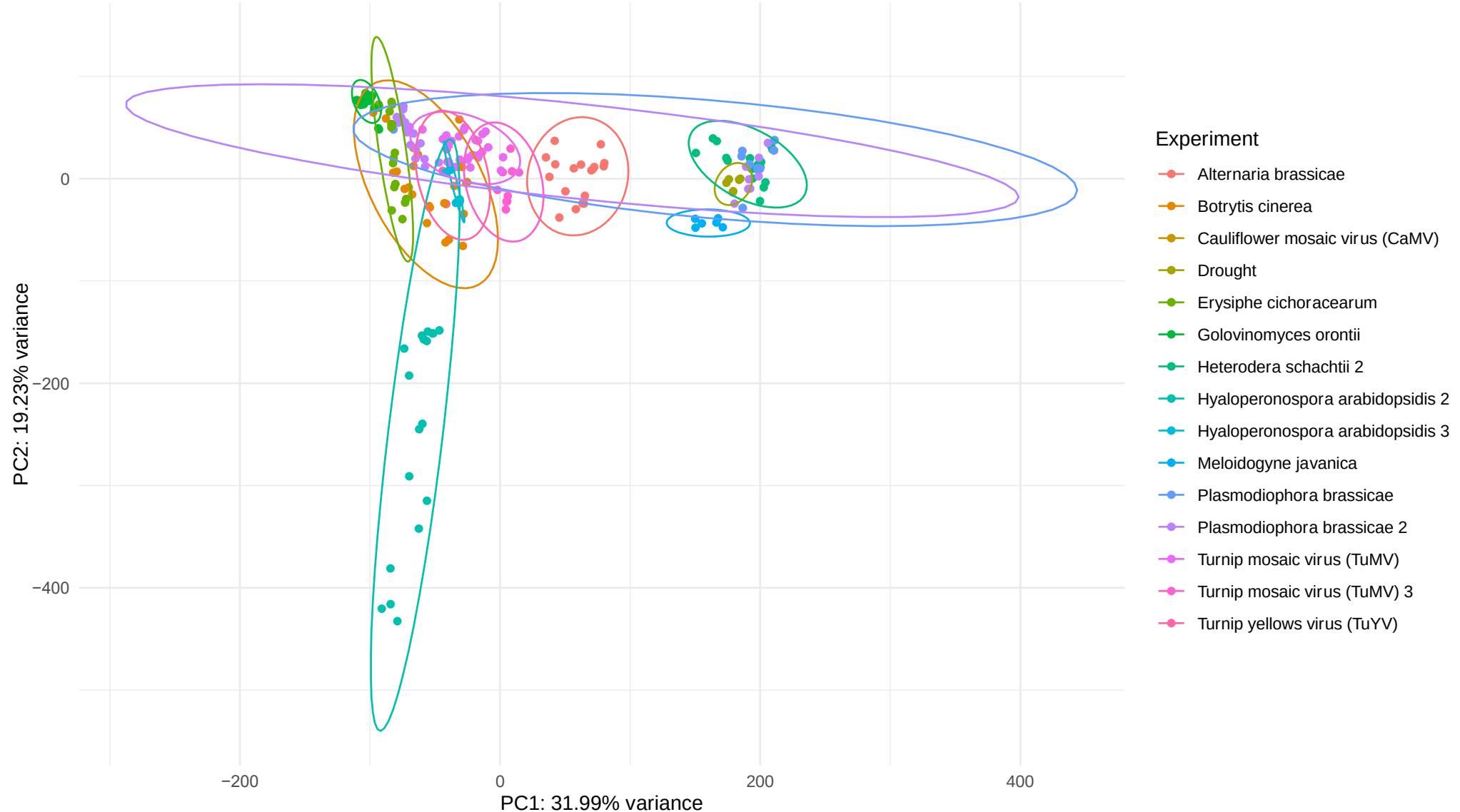


Kallisto: Alignment Scores for 283 samples



- Count table via tximport
- Created DESeqDataset
- VST (blind =FALSE)
- Filter genes with tpm? (like in the internship?)

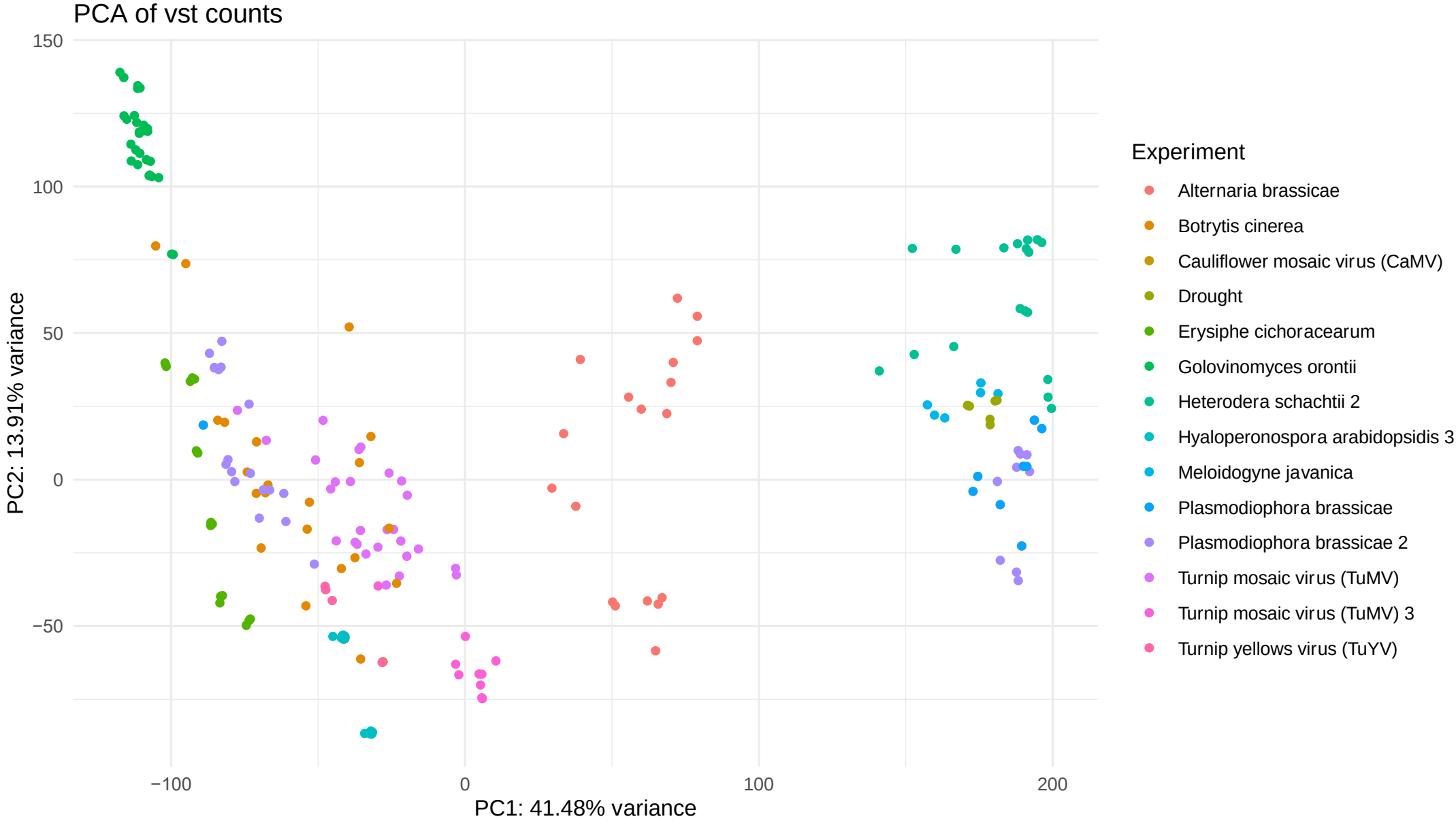
PCA of vst counts



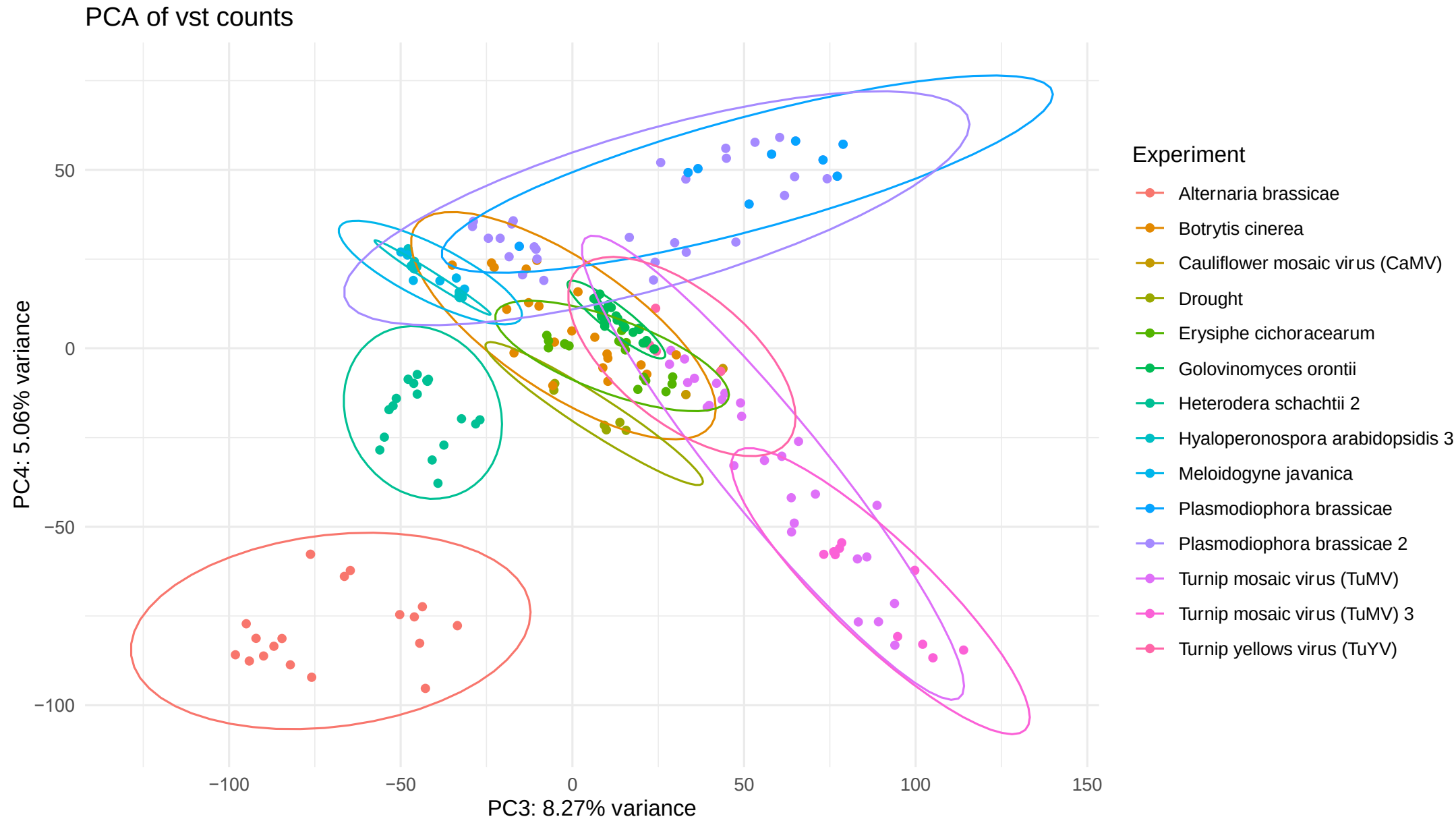
Outlier: Hpa2

- Already failed adapter content in multiqc after trimming
- Low number of aligned reads

Without Hpa2:



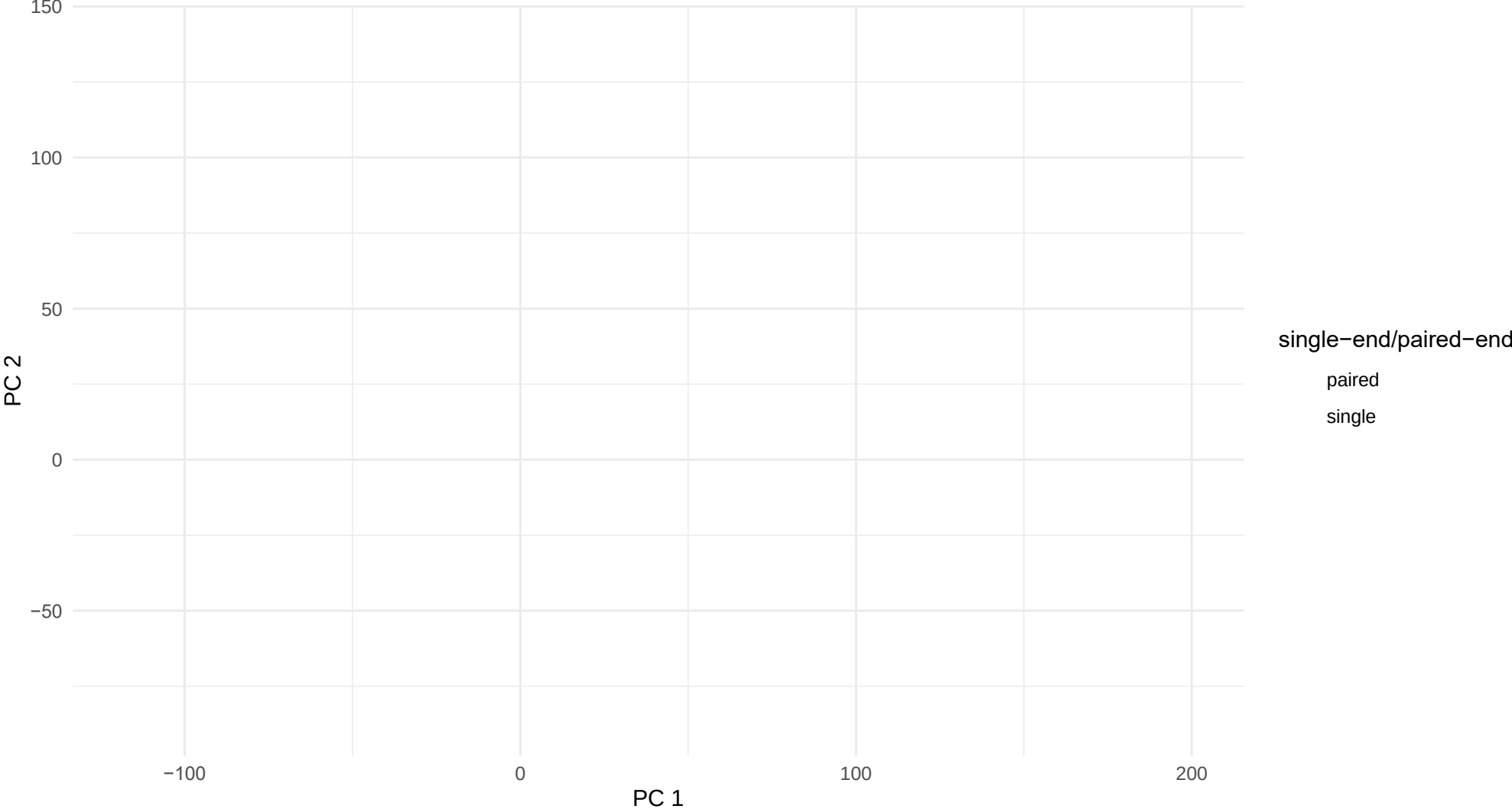
PC 3 and PC 4

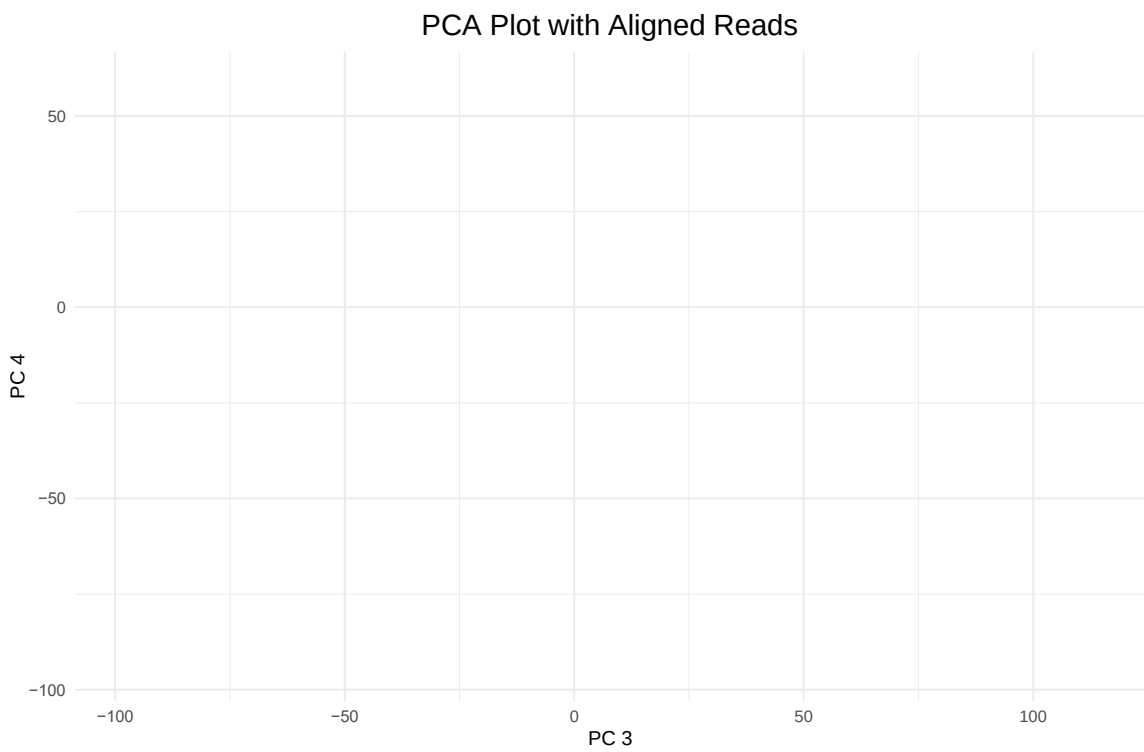


PCA with aligned reads per sample



PCA with vst counts





Aligned Reads

1e+07
2e+07
3e+07
4e+07
5e+07

Aligned Reads

5e+07
4e+07
3e+07
2e+07
1e+07

